Project - Stochastic Methods for Industrial Engineers

Submission: July 28, 2026

Instructions:

- The project is an open-ended assignment that lets you experiment and play from scratch with the concepts and algorithms we learned in class.
- In any case you prefer to go against the instructions, you will get full points (or more) given quality work that is not less in volume than the original instructions. Spend more time doing interesting things and learning and less time worrying about the instructions.
- The project sums up to 200 points, but you need to choose parts that amount to **at most 120** points to work on. Note that some parts build on others, or depend on the model you suggested. You can only choose the parts that make sense together.
- Each student submits the assignment independently based on their own work **alone**.
- You **cannot use AI** to solve the project and write your answers, but writing code is not considered part of the project.

1 Markov Chains

a) (20 points) **Markov Modeling**

   (a) (10 points) Choose a real-life random process that can be modeled as a Markov chain. Discuss why it is a good model, and also why it is a bad model.

   (b) (10 points) Is this chain irreducible? Aperiodic? Positive recurrent? Ergodic? Absorbing? Prove each of your assertions and discuss what assumptions were needed to make them.

b) (25 points) **Hidden Markov Models**

   (a) (10 points) Generate data based on your model from Q1. Design an HMM model for this data. Alternatively, suggest data from which a better Markov model can be fitted than the one you suggested in Q1. Fit a Markov model from data using HMM. Plot graphs that show the convergence of the EM algorithm.

   (b) (10 points) Show graphs that evaluate how good the HMM approximation from (a) is.

   (c) (5 points) Do you get different approximations when initializing the parameters differently? Why?

c) (30 points) **Markov Decision Process**

   (a) (10 points) Suggest how your Markovian model from Q1 can be generalized into an MDP, and formalize that MDP.

   (b) (10 points) Run value iteration to compute the optimal stationary policy for that MDP, and show plots that demonstrate the convergence.

   (c) (10 points) Run Q-learning and show the convergence to the same stationary policy from (b). How did you choose the step size? Discuss all other design choices as well.

d) (25 points) **Markov Chain Monte Carlo**

   (a) (10 points) Suggest a distribution that is hard to sample from. Explain the complexity of sampling from this distribution, or computing this distribution.

   (b) (10 points) Design a MCMC-based scheme that can sample from this distribution, and discuss your design choices.

   (c) (5 points) Show histograms that demonstrate that your MCMC achieves the target distribution.

2 Martingales

a) (40 points) **Martingale Modeling**

(a) (10 points) Choose a real-life random process that can be modeled as a martingale (supermartingale, submartingale, or almost). Discuss why it is a good model, and also why it is a bad model. Alternatively, choose a real life process that is not a martingale but can be analyzed using martingales (e.g., pattern matching from class).

(b) (10 points) Associate your process with a stopping time of interest and apply the optional stopping theorem to draw an interesting conclusion.

(c) (10 points) Prove that your process (or a related process) converges with probability 1. What does it converge to? Show that in simulations. Does it converge in L1? Explain.

(d) (10 points) Apply Azuma's inequality or Doob's inequality to draw interesting conclusions about your process.

b) (30 points) **Stochastic Gradient Descent**

(a) (10 points) Pick a stochastic gradient type scheme that was not studied in class. Explain the scheme and its motivation.

(b) (10 points) Prove convergence for that scheme based on tools we learned in class.

(c) (10 points) Plot the convergence of this scheme over many realizations.

c) (30 points) **Q-Learning**

(a) (10 points) Consider the Q-learning you ran in (c) of "Markov Chains". Does our proof from class explain why it converges? If yes, show that all assumptions hold. If not, what assumptions are violated?

(b) (20 points) Suggest ways to extend our proof from class. You will get more points the closer your suggestions are to a full proof.